

Complete `tcpdump` Command Examples Guide

Introduction

`tcpdump` is a powerful command-line packet analyzer used for:

- Network troubleshooting
- Security monitoring
- Packet capture analysis
- IDS/IPS investigations
- Protocol debugging
- Malware traffic analysis
- Incident response

It works on Linux, Unix, and macOS systems.

1. Installation

Install on Rocky Linux / CentOS / RHEL

```
sudo dnf install tcpdump -y
```

Install on Ubuntu / Debian

```
sudo apt install tcpdump -y
```

Install on Fedora

```
sudo dnf install tcpdump -y
```

Verify installation

```
tcpdump --version
```

2. Basic Packet Capture

Capture packets on default interface

```
sudo tcpdump
```

Capture packets on specific interface

```
sudo tcpdump -i eth0
```

List available interfaces

```
sudo tcpdump -D
```

Disable hostname resolution

```
sudo tcpdump -n
```

Disable hostname and port resolution

```
sudo tcpdump -nn
```

Show verbose output

```
sudo tcpdump -v
```

Show more verbose output

```
sudo tcpdump -vv
```

Maximum verbosity

```
sudo tcpdump -vvv
```

3. Packet Count and Capture Limits

Capture only 10 packets

```
sudo tcpdump -c 10
```

Capture 100 packets on interface

```
sudo tcpdump -i eth0 -c 100
```

Capture full packets

```
sudo tcpdump -s 0
```

Set snapshot length

```
sudo tcpdump -s 128
```

4. Save and Read Capture Files

Save packets to file

```
sudo tcpdump -i eth0 -w capture.pcap
```

Save with timestamped filename

```
sudo tcpdump -i eth0 -w capture-$(date +%F-%H%M).pcap
```

Read packets from file

```
tcpdump -r capture.pcap
```

Read capture with no name resolution

```
tcpdump -nn -r capture.pcap
```

Read only first 20 packets

```
tcpdump -r capture.pcap -c 20
```

5. Capture by Host

Capture traffic from specific host

```
sudo tcpdump host 192.168.1.10
```

Capture source host traffic

```
sudo tcpdump src host 192.168.1.10
```

Capture destination host traffic

```
sudo tcpdump dst host 192.168.1.20
```

Capture between two hosts

```
sudo tcpdump host 192.168.1.10 and 192.168.1.20
```

Exclude host traffic

```
sudo tcpdump not host 192.168.1.100
```

6. Capture by Port

Capture HTTP traffic

```
sudo tcpdump port 80
```

Capture HTTPS traffic

```
sudo tcpdump port 443
```

Capture SSH traffic

```
sudo tcpdump port 22
```

Capture DNS traffic

```
sudo tcpdump port 53
```

Capture MySQL traffic

```
sudo tcpdump port 3306
```

Capture source port

```
sudo tcpdump src port 53
```

Capture destination port

```
sudo tcpdump dst port 443
```

Capture multiple ports

```
sudo tcpdump 'port 80 or port 443'
```

Exclude port

```
sudo tcpdump not port 22
```

Capture port range

```
sudo tcpdump portrange 1-1024
```

7. Capture by Protocol

Capture TCP packets

```
sudo tcpdump tcp
```

Capture UDP packets

```
sudo tcpdump udp
```

Capture ICMP packets

```
sudo tcpdump icmp
```

Capture ARP traffic

```
sudo tcpdump arp
```

Capture IPv6 traffic

```
sudo tcpdump ip6
```

Capture only IP traffic

```
sudo tcpdump ip
```

8. Advanced Filtering

Capture HTTP traffic from one host

```
sudo tcpdump host 192.168.1.10 and port 80
```

Capture TCP traffic excluding SSH

```
sudo tcpdump tcp and not port 22
```

Capture traffic from subnet

```
sudo tcpdump net 192.168.1.0/24
```

Capture source subnet traffic

```
sudo tcpdump src net 10.0.0.0/8
```

Capture destination subnet traffic

```
sudo tcpdump dst net 172.16.0.0/16
```

Combine filters

```
sudo tcpdump '(port 80 or port 443) and host 192.168.1.50'
```

Exclude subnet

```
sudo tcpdump not net 192.168.1.0/24
```

9. View Packet Contents

Display packets in ASCII

```
sudo tcpdump -A
```

Display packets in HEX and ASCII

```
sudo tcpdump -X
```

Display packets in HEX

```
sudo tcpdump -xx
```

View full packet contents

```
sudo tcpdump -s 0 -X
```

Display link-level headers

```
sudo tcpdump -e
```

10. Capture HTTP Traffic

Capture HTTP GET requests

```
sudo tcpdump -A -s 0 'tcp port 80'
```


Capture HTTP POST requests

```
sudo tcpdump -A -s 0 'tcp port 80 and (((ip[2:2] - ((ip[0]&0xf)<2)) -  
((tcp[12]&0xf0)>>2)) != 0)'
```

Capture web browsing traffic

```
sudo tcpdump 'tcp port 80 or tcp port 443'
```

11. DNS Analysis

Capture DNS queries

```
sudo tcpdump port 53
```

Capture DNS traffic in detail

```
sudo tcpdump -vv port 53
```

Capture DNS queries from one host

```
sudo tcpdump host 192.168.1.5 and port 53
```

12. ICMP and Ping Analysis

Capture ping requests and replies

```
sudo tcpdump icmp
```

Capture only echo requests

```
sudo tcpdump 'icmp[icmptype] == icmp-echo'
```

Capture only echo replies

```
sudo tcpdump 'icmp[icmptype] == icmp-echoreply'
```

13. SSH Troubleshooting

Capture SSH traffic

```
sudo tcpdump port 22
```

Capture SSH traffic from one host

```
sudo tcpdump host 192.168.1.10 and port 22
```

Troubleshoot SSH connection resets

```
sudo tcpdump 'tcp port 22 and tcp[tcpflags] & tcp-rst != 0'
```

14. TCP Flag Analysis

Capture SYN packets

```
sudo tcpdump 'tcp[tcpflags] & tcp-syn != 0'
```

Capture SYN-ACK packets

```
sudo tcpdump 'tcp[tcpflags] & (tcp-syn|tcp-ack) != 0'
```

Capture FIN packets

```
sudo tcpdump 'tcp[tcpflags] & tcp-fin != 0'
```

Capture RST packets

```
sudo tcpdump 'tcp[tcpflags] & tcp-rst != 0'
```

Capture ACK packets

```
sudo tcpdump 'tcp[tcpflags] & tcp-ack != 0'
```

Capture PSH packets

```
sudo tcpdump 'tcp[tcpflags] & tcp-push != 0'
```

15. Wireless Traffic Capture

Capture wireless traffic

```
sudo tcpdump -i wlan0
```

Monitor mode capture

```
sudo tcpdump -i wlan0mon
```

Capture Wi-Fi management frames

```
sudo tcpdump -i wlan0 type mgt
```

16. VLAN Traffic

Capture VLAN traffic

```
sudo tcpdump vlan
```

Capture specific VLAN

```
sudo tcpdump vlan 100
```

17. Capture Packet Sizes

Capture packets larger than 1000 bytes

```
sudo tcpdump greater 1000
```

Capture packets smaller than 128 bytes

```
sudo tcpdump less 128
```

18. Time and Timestamp Options

Show human-readable timestamps

```
sudo tcpdump -tttt
```

Show packet arrival time

```
sudo tcpdump -tt
```

Show delta times between packets

```
sudo tcpdump -ttt
```

19. Rotating Capture Files

Rotate files every 10 MB

```
sudo tcpdump -i eth0 -C 10 -w capture.pcap
```

Rotate every hour

```
sudo tcpdump -i eth0 -G 3600 -w 'capture-%Y%m%d-%H%M%S.pcap'
```

Keep only 5 rotating files

```
sudo tcpdump -i eth0 -W 5 -C 10 -w capture.pcap
```

20. Security Monitoring Examples

Detect SYN flood attacks

```
sudo tcpdump 'tcp[tcpflags] & tcp-syn != 0 and not src net 192.168.1.0/24'
```

Detect port scanning

```
sudo tcpdump 'tcp[tcpflags] & tcp-syn != 0'
```

Monitor suspicious DNS traffic

```
sudo tcpdump port 53 and not host 8.8.8.8
```

Detect large outbound transfers

```
sudo tcpdump greater 5000
```

Monitor failed connections

```
sudo tcpdump 'tcp[tcpflags] & tcp-rst != 0'
```

21. Real-World Troubleshooting Examples

Troubleshoot web server

```
sudo tcpdump -i eth0 host 192.168.1.100 and '(port 80 or port 443)'
```

Troubleshoot database connectivity

```
sudo tcpdump port 3306
```

Troubleshoot DNS resolution

```
sudo tcpdump -i eth0 port 53
```

Troubleshoot SMTP mail flow

```
sudo tcpdump port 25
```

Troubleshoot NTP synchronization

```
sudo tcpdump port 123
```

22. Useful Output Options

Print packet numbers

```
sudo tcpdump -v
```

Do not print packet data

```
sudo tcpdump -q
```

Quick output mode

```
sudo tcpdump -t
```

Line-buffered output

```
sudo tcpdump -l
```

23. Combining with Other Commands

Save and compress capture

```
sudo tcpdump -i eth0 -w - | gzip > capture.pcap.gz
```

Analyze live traffic with grep

```
sudo tcpdump -A port 80 | grep "Host:"
```

Capture and analyze with Wireshark later

```
sudo tcpdump -i eth0 -w traffic.pcap
```

24. Performance Optimization

Use larger buffer size

```
sudo tcpdump -B 4096
```

Capture only headers

```
sudo tcpdump -s 64
```

Reduce CPU usage

```
sudo tcpdump -q
```

25. Common Tcpdump Filters Reference

Filter	Meaning
host 1.1.1.1	Traffic to/from host
src host 1.1.1.1	Source host
dst host 1.1.1.1	Destination host
port 80	Port 80 traffic
src port 53	Source port
dst port 443	Destination port
tcp	TCP traffic
udp	UDP traffic
icmp	ICMP traffic
arp	ARP traffic
net 192.168.1.0/24	Network traffic
not	Negation
and	Logical AND

Filter	Meaning
or	Logical OR

26. Best Practices

1. Use `-nn` for faster captures.
 2. Save captures to `.pcap` files for analysis.
 3. Use filters to reduce unnecessary traffic.
 4. Avoid long captures on busy systems.
 5. Rotate capture files in production.
 6. Use snapshot lengths for performance.
 7. Secure capture files because they may contain sensitive data.
 8. Analyze captures with Wireshark when needed.
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27. Common Tcpdump Workflow

Step 1: Identify interface

```
sudo tcpdump -D
```

Step 2: Start capture

```
sudo tcpdump -i eth0 -nn -w troubleshooting.pcap
```

Step 3: Reproduce issue

Perform the network activity.

Step 4: Stop capture

Press:

```
CTRL+C
```

Step 5: Analyze capture

```
tcpdump -r troubleshooting.pcap
```

28. Troubleshooting Tips

Permission denied

Run tcpdump with sudo:

```
sudo tcpdump
```

No packets captured

Verify interface:

```
ip addr
```

Capture dropped packets

Increase buffer size:

```
sudo tcpdump -B 8192
```

High CPU usage

Use filters and reduce verbosity.

Conclusion

This guide covers:

- Basic packet captures
- Advanced filtering
- Protocol analysis
- Security monitoring

- Incident response
- Troubleshooting workflows
- Real-world tcpdump examples

`tcpdump` is one of the most essential tools for:

- Network engineers
- Linux administrators
- Security analysts
- SOC teams
- Incident responders
- Penetration testers
- IDS/IPS engineers